DLS Buddy — Quickstart Guide

Applies to v1.0.0 and later

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Contents

1. Introduction

DLS Buddy analyzes **dynamic (DLS)** and **static (SLS)** light scattering data.

This guide covers the common path — load a file, confirm its parameters, run a DLS or SLS analysis, and export the result. For every supported format, the Cross-Sample and Utilities tools, and the full parameter reference, see the **User-Manual**. For the equations and the literature behind each method, see the **Theory-and-Equations-Guide**.

The window is a left **sidebar** (your loaded data, auto-grouped into samples) plus tabbed modules. The two you need first are **Data** (confirm parameters) and then **DLS** or **SLS** (analyze).

2. Launching the Program

Double-click the launcher for your operating system:

- **Windows** — `Launch DLS Buddy (Windows).bat`
- **macOS** — `Launch DLS Buddy (MacOS).command` (on the first run, macOS Gatekeeper may refuse an unsigned script — right-click it → **Open** → confirm once, and it opens normally thereafter)

Or run it from the repository root:

```
python -m gui.main
```

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3. The Workflow at a Glance

Almost every analysis follows the same short path:

Launch → Load → Confirm parameters + Update → open DLS or SLS → pick a method and Run → read the result → Export.

1. **Launch** the program.
2. **Load** your data — **Load DLS correlogram...** or **Load SLS intensities...**. Files are auto-detected and grouped into **samples** in the sidebar.
3. **Confirm the parameters** on the **Data** tab, then click **Update** to commit.
4. **Open the analysis tab: DLS** for size and $R\mathrm{h}$, **SLS** for $M\mathrm{w}$, $R\mathrm{g}$, and A_2.
5. **Pick a method and Run**, then read the result in the table and plot.
6. **Export** with **Export CSV...**.

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4. Loading Data

Click **Load DLS correlogram...** or **Load SLS intensities...** and pick one or more files.

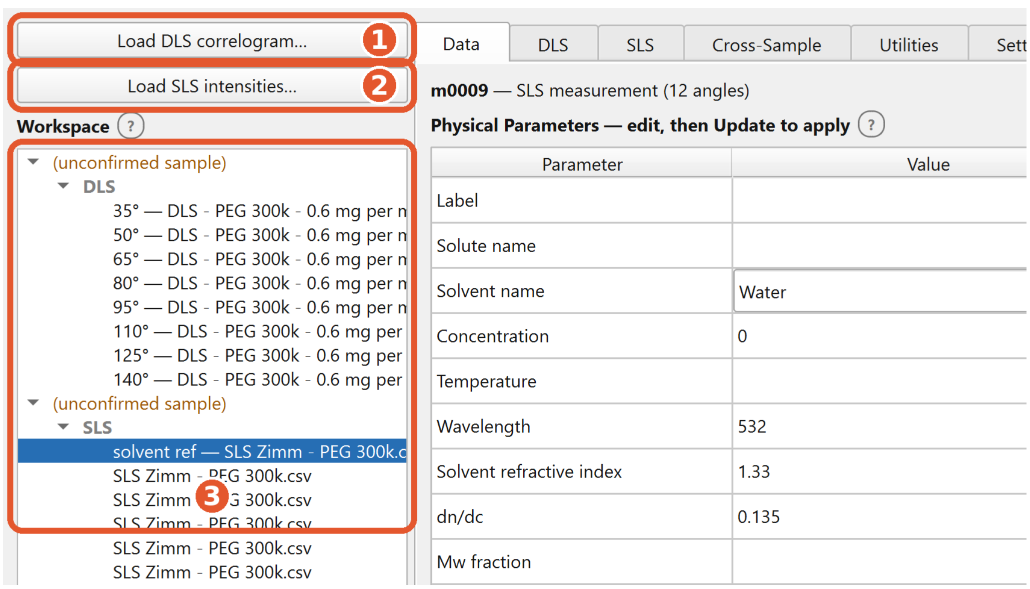

Figure 1: Loading data: click **Load DLS correlogram...** (1) or **Load SLS intensities...** (2). The loaded measurements appear in the Workspace sidebar (3), auto-grouped into a sample — here one file has loaded eight scattering angles — and their parameters populate the **Data** tab, ready to confirm.

The parser pre-fills whatever parameters the file contains. You can override any of the pre-filled values on the **Data** tab.

> **Note.** Some formats carry no metadata, so you enter every parameter at the confirmation step. For the full list of formats and exactly what each one pre-fills, see the **User-Manual**.

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5. Confirming Parameters

The **Data** tab shows every parameter for the selected measurement. Fill in any blank field and correct any pre-filled value, then click **Update** to commit.

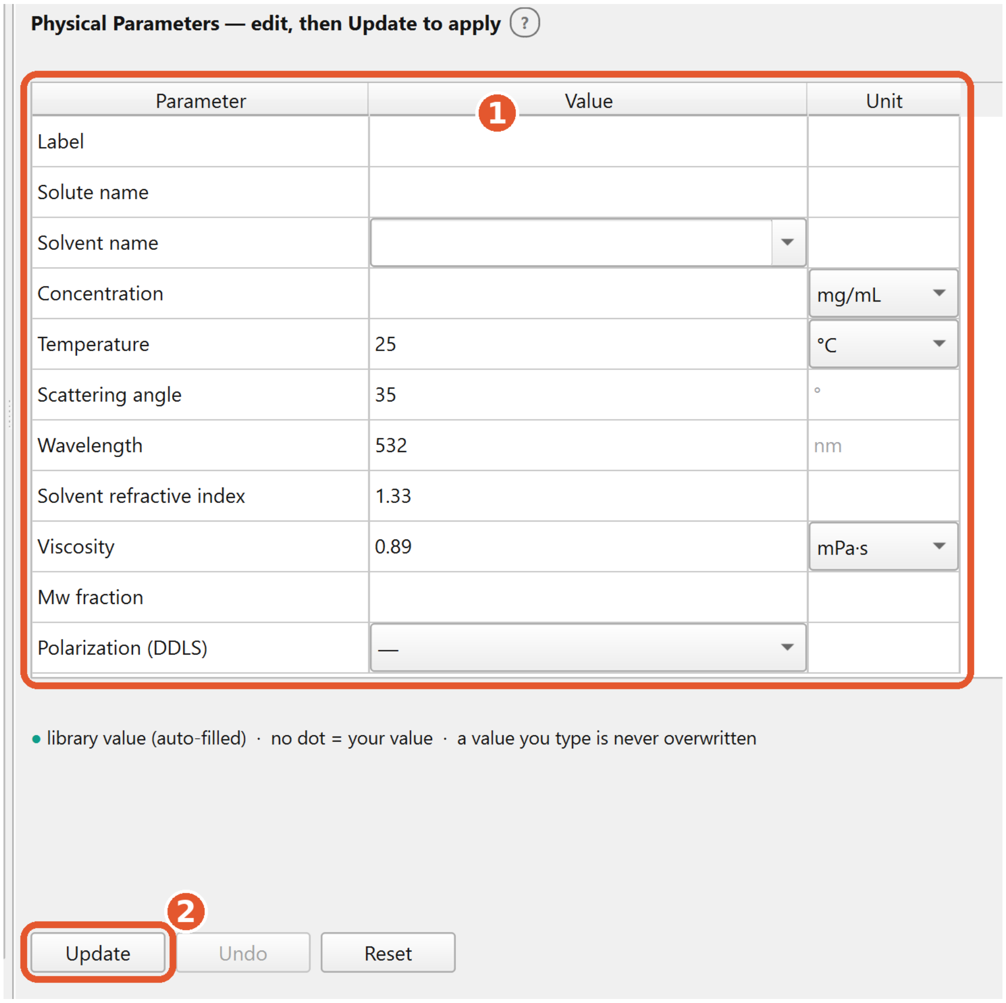

Figure 2: On the **Data** tab, edit each parameter and choose its unit in the dropdown (1), then press **Update** (2) to commit. A green dot marks a value the solvent library auto-filled; a value you type yourself is never overwritten.

- **Identity and optics are entered once per sample** — polymer/sample name, solvent, temperature, refractive index, wavelength, and scattering angle. Press **Enter** to jump to the next field.
- Each parameter has a **Unit** column: a dropdown where a quantity has several natural units (switching units re-displays the value; the stored physical value is unchanged), or a fixed label otherwise.
- For common solvents, picking the **Solvent name** from the dropdown fills in the refractive index and viscosity from the sample's temperature and wavelength, each marked with a small **teal dot**. A value you type by hand is never overwritten and shows no dot.
- **Concentration** is required for SLS and for the DLS concentration series; single cumulant / exponential / distribution size analysis does not use it.

- **To set a value on several measurements at once**, Ctrl- or Shift-click them in the sidebar, edit the focused one, and press **Update** — your edits apply to every highlighted measurement.

Analysis always runs on the **committed** values. If you change a parameter that fed a fit already on screen, the result is kept but flagged stale until you re-run it. The full unit table and the solvent library are in the **User-Manual**.

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6. DLS

The **DLS** tab extracts particle dynamics — decay rates, diffusion coefficients, and hydrodynamic radii — from a measured $g_2(\tau)-1$.

Tick the measurement(s) to analyze in the **Measurements to plot** checklist. This checklist is the sole driver of what gets fit; selecting a measurement in the sidebar only navigates. The raw correlogram appears as soon as you tick it.

Then choose one route:

- **Correlogram** — parametric fits. The **cumulant** fit gives the intensity-weighted (z-average) Rh and the polydispersity index (PDI); single/double exponential and KWW fits are also here. Set the cumulant order in this sub-tab. A PDI above 0.3 is flagged as unreliable.
- **Distribution — CONTIN / NNLS / Lognormal**. These recover a full distribution of sizes without assuming a shape: NNLS resolves well-separated peaks but is spiky, CONTIN is smoother and generally better for broad or real data, and Lognormal is a robust single-mode default. Resolved peaks are labeled on the plot and listed in the **Peak results** table.

Set the **delay window** and **baseline region** on the Correlogram sub-tab — type the values or drag the markers on the plot (green carets along the top are the window; purple carets along the bottom are the baseline). They apply to every ticked measurement.

Click **Run** to fit, and read the result in the table and plot. A single cumulant or distribution peak is an **apparent** Rh at one angle and concentration.

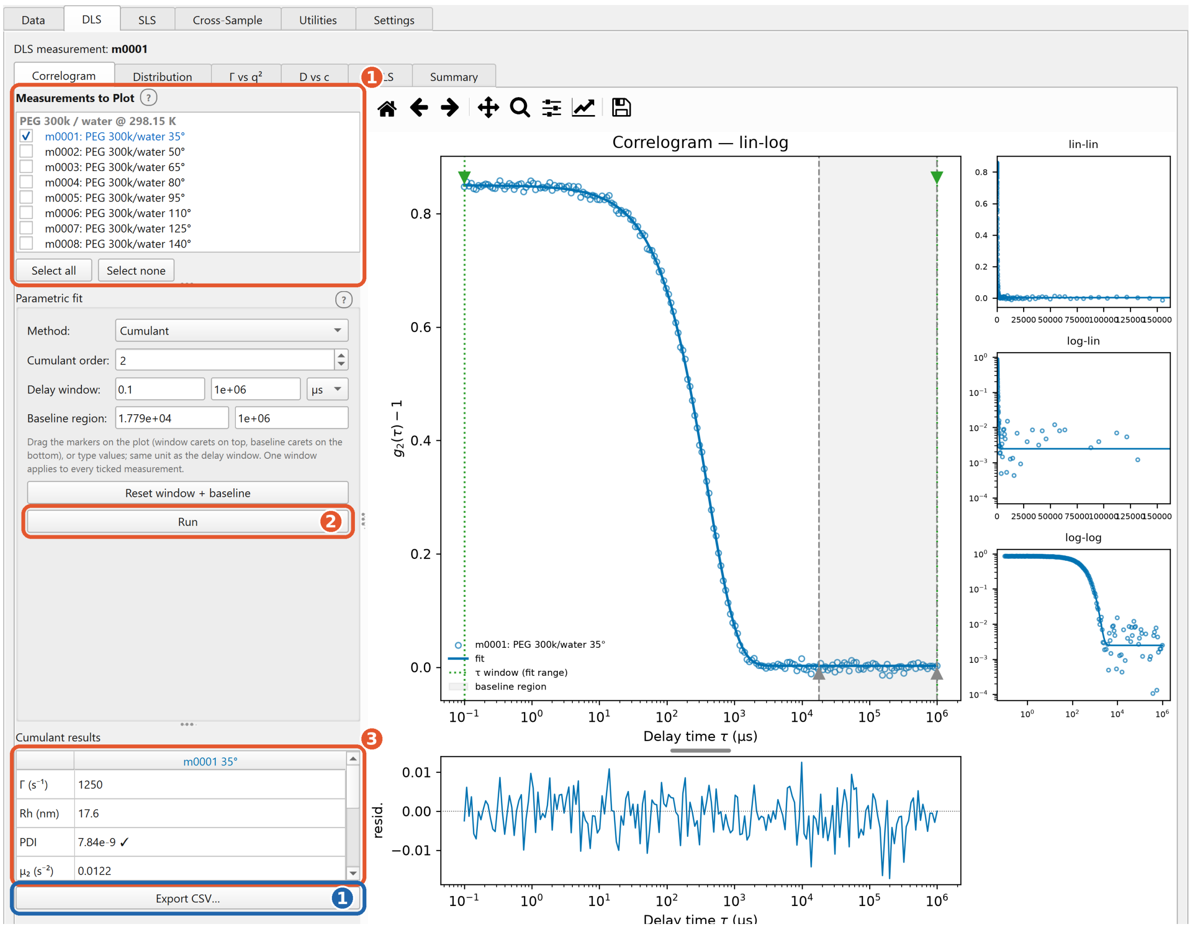

Figure 3: The DLS **Correlogram** sub-tab. *Orange:* tick the measurement(s) to plot (1) and press **Run** (2); read Rh and PDI in the results table (3). The correlogram and its cumulant fit fill the plot on the right. *Blue:* **Export CSV...** writes the result on screen.

> **Tip.** Fitting many heavy distributions (especially CONTIN) at once can be slow — tick fewer items if it feels sluggish.

The multi-angle (**Γ vs q**^{2}), concentration-series (**D vs c**), and depolarized (**DDLS**) analyses, and the workspace-wide **Summary**, are covered in the **User-Manual**.

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7. SLS

The **SLS** tab extracts the weight-average molecular weight Mw, the radius of gyration Rg, and the second virial coefficient A_2. Pick the sample from the **SLS sample** dropdown at the top; the whole set — solvent reference plus the concentration series — is analyzed together.

1. **Set the calibration.** The program computes its own calibration constant kc from one calibrant measurement — the intensity of a standard liquid (usually toluene), the angle, and that standard's geometry-aware Rayleigh ratio. Any value from a data file is kept for reference only and not used. If you supply no calibration, the analysis still runs on an arbitrary scale, flagged **uncalibrated**: then Mw and the absolute A_2 are unreliable, but Rg and the

calibration-free product $2A_2M\mathrm{w}$ remain valid.

2. **Choose a method.** **Zimm** or **Berry** double extrapolation combines several concentrations and angles, extrapolates to zero angle and zero concentration, and returns $M\mathrm{w}$, $R\mathrm{g}$, and A_2 at once — a **thermodynamic** result. (Berry linearizes the high-$qR\mathrm{g}$ regime of large $M\mathrm{w}$ better than Zimm; the program reports which it used.) A single-concentration **Debye** fit gives only **apparent** values.
3. **Mask any bad points** (click to hide an angle, a concentration, or a single point), then click **Run** and read the result. Flags mark apparent vs thermodynamic and calibrated vs uncalibrated.

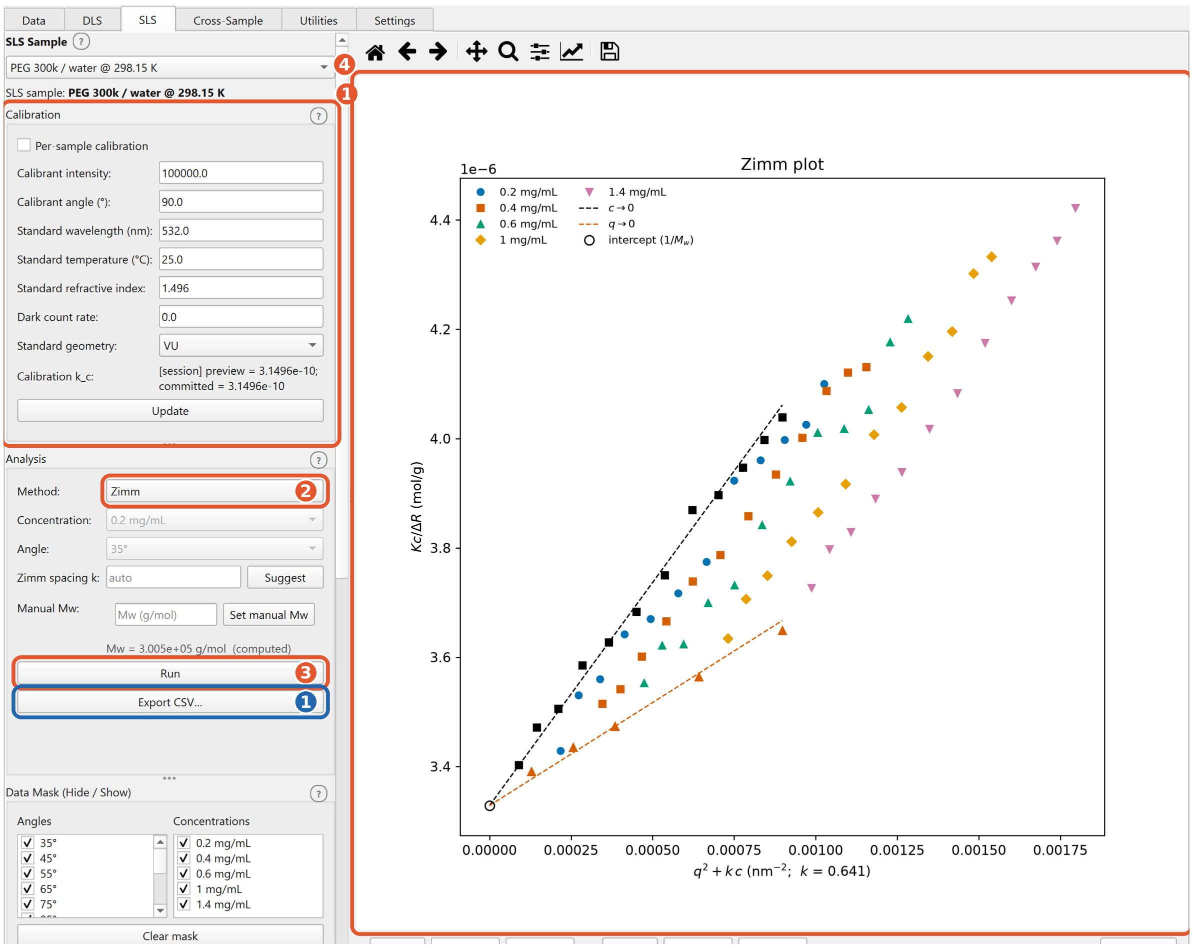

Figure 4: The **SLS** tab. *Orange:* set the calibration (1), choose the method — **Zimm** here (2) — and press **Run** (3); the Zimm double-extrapolation plot and $M\mathrm{w}$ / $R\mathrm{g}$ / A_2 appear (4). *Blue:* **Export CSV...** writes the result.

Guinier, single-angle $M\mathrm{w}$, the calibration-free A_2 route, the depolarization correction, and the manual-$M\mathrm{w}$ override are covered in the **User-Manual**.

8. Exporting Results

Every analysis view has an **Export CSV...** button that writes the result currently on screen to a CSV that OriginLab imports directly. Each file carries the four Origin label rows (Long Name, Units, Comments, Parameters), so axis labels and units populate on import. An uncalibrated result writes "uncalibrated, arbitrary scale" into the **Comments** cell of the affected columns only, without changing the row count, so the Origin import stays intact.

Data-quality flags are interface overlays, not part of the figure — a saved plot image is always clean.